

PROJECT ASSIGNMENT

Bayesian Data Analysis

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INSTRUCTIONS

- Establish groups of the size of about 3 students
- Make a report in question-answer style
- Describe the flow of your procedures and the reasoning behind them.
- Limit your report to 20 pages (excluding title page, but including tables and figures)
- **Deadline of report is January 5, 2026 (at 23h59)**
- Send also your programs so that we can check that the programs really work!
- Submit report and R program via Blackboard

PART 1

In this part, you will have to do a Bayesian analyses with MCMC inference in `NIMBLE` or `jags`. In each step, do assess and report on the convergence. Submit runnable code. Use clear section headings corresponding to each of the questions.

Data description

The German socioeconomic panel study data was taken from the first twelve annual waves (1984 through 1995) of the German Socioeconomic Panel (GSOEP) which surveys a representative sample of East and West German households. The data provide detailed information on the utilization of health care facilities, characteristics of current employment, and the insurance schemes under which individuals are covered. We consider a random sample of 100 individuals aged 25 through 65 from the West German subsample and of German nationality, which participated throughout all years.

The dataset contains the following variables:

- `id`: person - identification number
- `female`: female = 1; male = 0
- `year`: calendar year of the observation
- `age`: age in years
- `hsat`: health satisfaction, coded 0 (low) - 10 (high)
- `handdum`: handicapped = 1; otherwise = 0
- `handper`: degree of handicap in percent (0 - 100)
- `hhninc`: household nominal monthly net income in German marks / 1000
- `hhkids`: children under age 16 in the household = 1; otherwise = 0
- `educ`: years of schooling
- `married`: married = 1; otherwise = 0
- `working`: employed = 1; otherwise = 0
- `docvis`: number of doctor visits in last three months
- `hospviz`: number of hospital visits in last calendar year
- `public`: insured in public health insurance = 1; otherwise = 0

You can read in the data in R using the command `readRDS("GSPS.RData")`

Questions

1. Model 1: Build and fit a logistic mixed effects model to investigate the employment status over time. Use a random intercept to account for the individual heterogeneity. No other covariates have to be taken into account in this model. Describe the model that you use (likelihood and prior specification). Implement the Bayesian model, give appropriate inference summaries and interpret results.
2. Model 2-3: Investigate which of the covariates influence the employment status. To study this, implement automatic variable selection via ridge (Gaussian) and lasso (Laplace via normal-exponential mixture) priors. Look up in literature how a *ridge* model and *lasso* model can be cast into a Bayesian hierarchical regression model. In the report, explain the Bayesian *ridge* and *lasso* model specification (likelihood and priors). Implement the models, give appropriate inference summaries and interpret results. Compare the variable selection results from both models.
3. Model comparison: Look up in literature how WAIC is defined, and explain briefly. Compute WAIC and DIC. Discuss which model is preferred and whether differences are practically important.
4. Model evaluation: Generate replicated data $\tilde{y}$ from posterior draws. Compare distributions and at least one discrepancy (e.g. SD, skewness, maximum).

PART 2

Markov chain Monte Carlo

An experiment is conducted in a laboratory to assess the strength of infection of $n = 8$ different artificial viruses on microorganisms. The number of infected organisms (y_i) during an exposure period (t_i) to a virus (v_i) is given below.

Virus (v_i)	v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8
Infections (y_i)	4	1	5	14	3	19	7	6
Exposure period (t_i)	95	16	63	126	6	32	16	19

Assume that the number of infections generated by the i th virus follows a Poisson distribution with parameter $\lambda_i > 0$, $i \in \{1, \dots, 8\}$, so that for an exposure time t_i , the number of infections y_i is Poisson distributed $y_i \sim \mathcal{P}(\lambda_i t_i)$ with mean $\mathbb{E}(y_i) = \lambda_i t_i$. In addition, assume the following Gamma prior distributions $\lambda_i \sim_{i.i.d.} \mathcal{G}(\alpha, \beta)$ for $i = 1, \dots, n$ with $\alpha = 1.8$ and $\beta \sim \mathcal{G}(\gamma, \delta)$ with $\gamma = 0.01$ and $\delta = 1$, where $\mathcal{G}$ denotes a gamma distribution with the shape-rate parameterization.

1. Using Bayes' theorem compute the joint posterior distribution $p(\boldsymbol{\lambda}, \beta | \mathcal{D})$, where $\mathcal{D}$ denotes the observed data and $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_8)^\top$.

Random-walk Metropolis algorithm

2. Based on the joint posterior $p(\boldsymbol{\lambda}, \beta | \mathcal{D})$, compute the log posterior $\log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})$, where $\tilde{\boldsymbol{\lambda}} = (\tilde{\lambda}_1, \dots, \tilde{\lambda}_8)^\top$ with $\tilde{\lambda}_i = \log(\lambda_i)$ and $\tilde{\beta} = \log(\beta)$.
3. Obtain analytically the gradient $\nabla \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})$ and Hessian matrix $\nabla^2 \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})$:

$$\nabla \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D}) = \left(\frac{\partial \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\lambda}_1}, \dots, \frac{\partial \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\lambda}_8}, \frac{\partial \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\beta}} \right)^\top,$$

$$\nabla^2 \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D}) = \begin{pmatrix} \frac{\partial^2 \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\lambda}_1^2} & \frac{\partial^2 \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\lambda}_1 \partial \tilde{\beta}} \\ \frac{\partial^2 \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\beta} \partial \tilde{\lambda}_1} & \frac{\partial^2 \log p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})}{\partial \tilde{\beta}^2} \end{pmatrix}.$$

4. Using the gradient and Hessian matrix obtained in the previous step, implement a Newton-Raphson (NR) algorithm in R software to find a maximizer of $p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})$ (use the zero vector as a starting point in your NR algorithm). Denote by $\boldsymbol{\theta}_{NR}^* = (\tilde{\boldsymbol{\lambda}}^*, \tilde{\beta}^*)^\top$ the maximizer obtained after convergence of your algorithm and report $\boldsymbol{\theta}_{NR}^*$ by rounding it to five digits after the decimal point.
5. Compute a Laplace approximation to $p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})$ around $\boldsymbol{\theta}_{NR}^*$. Report the covariance matrix Σ^* of the Laplace approximation (Note: round your results to five digits after the decimal point).
6. In R, write a random-walk Metropolis algorithm to explore the joint posterior $p(\tilde{\boldsymbol{\lambda}}, \tilde{\beta} | \mathcal{D})$ using a Gaussian proposal with covariance matrix $\tilde{\Sigma} = c\Sigma^*$. Use a chain of length $M = 35\,000$ (including a burn-in of size 5\,000) and tune c to (approximately) reach the optimal acceptance rate of 23%. Please specify *set.seed(1995)* for the sake of reproducibility of your results.
7. Based on the posterior mean of your generated chain, provide a point estimate and 95% quantile-based credible interval for $\mathbb{E}(y_4)$, i.e. the average number of infections generated by virus v_4 for $t_4 = 126$ minutes. (Round your results to the nearest integer).
8. Using your generated chain, estimate the probability $\mathbb{P}(\lambda_4 > 0.11)$.

Gibbs sampling

9. Based on the joint posterior $p(\boldsymbol{\lambda}, \beta | \mathcal{D})$ obtained in Question 1, compute the following conditional posterior distributions $p(\lambda_i | \boldsymbol{\lambda}_{-i}, \beta, \mathcal{D})$ for $i = 1, \dots, 8$ and $p(\beta | \boldsymbol{\lambda}, \mathcal{D})$. To what well-known parametric family do these conditional posterior distributions belong?
10. Based on the full conditionals obtained in the previous step, write a pseudo-code to obtain a random sample of total size 35 000 (including a burn-in of size 5 000) from $p(\boldsymbol{\lambda}, \beta | \mathcal{D})$ using the Gibbs sampler.
11. Using R and without relying on external software/packages (except the *coda* package), write the Gibbs sampler based on your pseudo-code. Please specify *set.seed(2025)* at the beginning of your code for reproducibility of your results. What is your acceptance rate?
12. Use the Geweke convergence diagnostic test and the Heidelberger-Welch stationarity test on the generated chains and interpret your results.
13. Based on the posterior mean of your generated chain, provide a point estimate and 95% quantile-based credible interval for $\mathbb{E}(y_6)$, i.e. the average number of infections generated by virus v_6 for $t_6 = 32$ minutes. (Round your results to the nearest integer).
14. Using your generated chain, estimate the probability $\mathbb{P}(\lambda_6 > 0.53)$.